

Is the 2-space  $\mathbb{R}^2$  a subset/subspace of the 3-space  $\mathbb{R}^3$ ?

Answer:

No,  $\mathbb{R}^2$  is not a subset of  $\mathbb{R}^3$  because they have different dimensions. As mentioned in the course materials, different dimensional vector spaces like  $\mathbb{R}^2$  and  $\mathbb{R}^3$  are disjoint since they consist of different types of vectors — 2-component vectors for  $\mathbb{R}^2$  and 3-component vectors for  $\mathbb{R}^3$ .

However,  $\mathbb{R}^2$  can be seen as “isomorphic” to a subspace of  $\mathbb{R}^3$  specifically the xy-plane where each 2-vector  $(x, y)$  in  $\mathbb{R}^2$  corresponds to the 3-vector  $(x, y, 0)$  in  $\mathbb{R}^3$ . This subspace is spanned by two vectors and has dimension 2, but as standard sets,  $\mathbb{R}^2$  is not a subset of  $\mathbb{R}^3$  unless such an embedding is explicitly considered.